Every finite universal tree of the SEM forcing has a valid leaf

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Editorial note. The counted unit is the article's own Lemma lem:sem-wf-valid-leaf at source lines 880-882 together with the proof environment that begins on the next line (883-889); the pairing is the ordinary adjacent one. The remark that a maximal universal tree need not be M-finite at source line 878 precedes the statement and is not retained.

Editorial note. The retained context at source lines 819-829 and 853-861 is the author's own definitions of nodes, of validity, of trees and of universal trees; they are the ambient definitions of this section and are shared with the sibling cards of this lane.

Editorial note. The retained preamble reproduces verbatim the author's document class at source line 2, his theorem declarations at source lines 16-25, his macro definitions at source lines 46-88 and 91, and his natural-operation macros at source lines 104-107, and it retains mathabx and cleveref; the stix-dependent LS2 block and the inputenc load with the utf8x option are not retained, the former because the host lacks stix and the latter because it fights xelatex. Nothing was renamed or redefined.

Editorial note. No retained line contains a citation or a cross-reference, so the material carries no bibliography and needs no navigation pointer.

Editorial note. The article's conservation theorems, its COH and WKL lemmas and the other lemmas of the SEM section are not counted here and are not used as premises.

Allowed standard prerequisites (editorial).

- Nodes, validity and trees of the SEM forcing. A (sigma, a)-node is an M-finite sequence of pairs of a bit and an M-finite R-transitive subset of X_a, comparable according to the bits; it is valid when each block lies in the corresponding side of the colouring; a (sigma, a)-tree is a set of nodes closed downward under prefixes and it is universal when every non-leaf node can be extended on one of the two sides of any finite partition of a sufficiently long initial segment; the author's own definitions are retained at source lines 819-829 and 853-861.. Fixes all three objects of the counted

lemma, namely the finite universal tree, its leaves and the notion of validity that the leaf must satisfy.

- The colouring sides A_0(R) and A_1(R) and the scheme Delta^0_2. For the stable colouring R the sets A_0(R) and A_1(R) are the two sides of the partition of the integers induced by R; the scheme Delta^0_2(Y) induction, equivalent to collection for Sigma^0_2, is available in the models of this section.. The proof proceeds by Delta^0_2 induction along the finite tree and applies universality to the restriction of the two sides to an initial segment, whose M-finiteness is exactly what that scheme provides.

Definition 0.1. *Fix a condition* (σ, a).

(1) *A* (σ, a)-node *is an M-finite sequence of the form*

$$\nu = \langle (i_0, \rho_0), (i_1, \rho_1), \dots (i_{q-1}, \rho_{q-1}) \rangle$$

such that for every $s < q$, $i_s < 2$, $\rho_s \subseteq X_a$ *is a non-empty M-finite R-transitive set with* $\min \rho_s > \max \rho_{s-1}$ *(letting* $\rho_{-1} = \sigma$*), and for every* $s < t < q$, *if* $i_s = 0$ *then* $\rho_s \to_R \rho_t$, *and if* $i_s = 1$ *then* $\rho_t \to_R \rho_s$. *We then write*

$$\pi(\nu) = \sigma \cup \rho_0 \cup \dots \cup \rho_{q-1}$$

(2) *This* (σ, a)*-node is* valid *if for every* $s < q$, $\rho_s \subseteq A_{i_s}(R)$

(3) *A* (σ, a)*-node* ν satisfies *a* $\Sigma^0_1(\mathcal{M})$*-formula* $\varphi(G, x) \equiv \exists i \psi(G\restriction i, x)$ *if* $\psi(\pi(\nu), |\nu| - 1)$ *holds.*

Definition 0.2. *Fix a condition* (σ, a).

(1) *A* (σ, a)-tree *is a set of* (σ, a)*-nodes closed downard under the prefix relation.*

(2) *A* (σ, a)*-tree E is* universal *if for every* (σ, a)*-node* ν *which is not a leaf, there is some* $r \in M$ *such that for every M-finite 2-partition* $A_0 \sqcup A_1 = \{0, \dots, r-1\}$, *there is a side* $i < 2$ *and an M-finite R-transitive* $\rho \subseteq A_i \cap X_a$ *such that* $\nu \cdot \langle i, \rho \rangle \in E$.

(3) *A* (σ, a)*-tree E satisfies a* $\Sigma^0_1(\mathcal{M})$ *formula if all its nodes satisfies it.*

Lemma 0.3. *Every M-finite universal* (σ, a)*-tree E has a valid leaf.*

Proof. We prove it by $\mathsf{I}\Delta^0_2$ induction. The root $\langle\rangle$ is vacuously valid. Assume $\nu \in E$ is a valid non-leaf. By universality of E, there is some $r \in M$ such that for every M-finite 2-partition $A_0 \sqcup A_1 = \{0, \dots, r-1\}$, there is a side $i < 2$ and an M-finite R-transitive $\rho \subseteq A_i \cap X_a$ such that $\nu \cdot \langle i, \rho \rangle \in E$. By $\mathsf{BC}\Delta^0_2(Y)$ (which is equivalent to $\mathsf{B}\Sigma^0_2(Y)$), the following partition is M-finite:

$$A_0(R)\restriction_r \sqcup A_1(R)\restriction_r = \{0, \dots, r-1\}$$

Thus, there exists some $i < 2$ and some $\rho \subseteq A_i(R)$ such that $\nu \cdot \langle i, \rho \rangle \in E$. By choice of i and ρ, the node $\nu \cdot \langle i, \rho \rangle$ is valid. □